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"Do not let your difficulties fill you with anxiety, after all it is only in the darkest nights that stars shine more brightly."

"Life consists of two days, one for you one against you.
(Hazrat Ali Ibn Abu-Talib A.S)

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MTH 304:

MECHANICS

Notes

Lectures No (33-39)

Written By

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0302-4865858

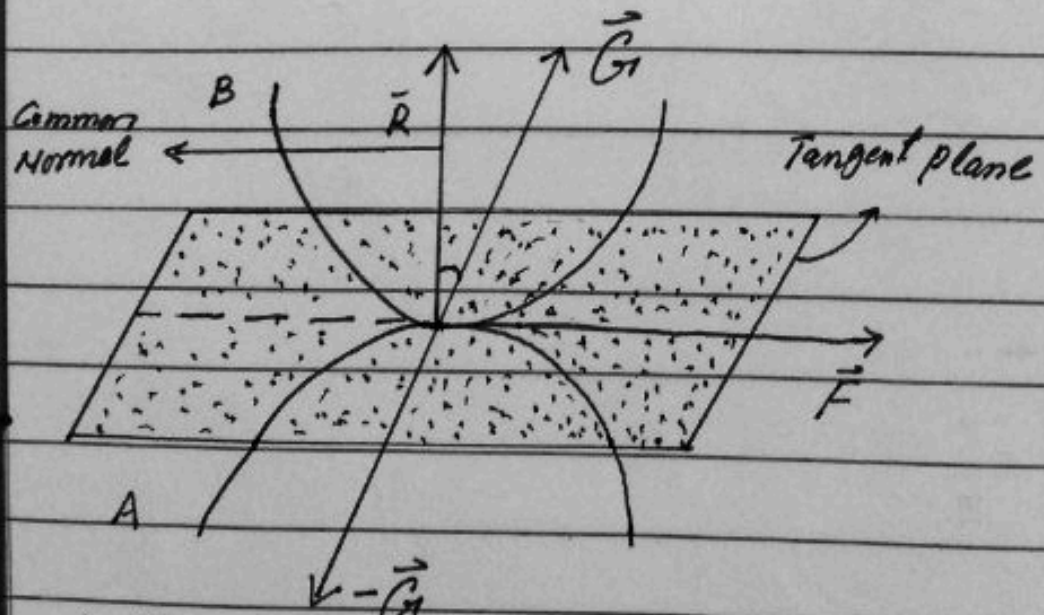
MTH:304

"Friction"

"It is Common observation That" Whenever two bodies are in Contact with each other, each exerts a force on the other. Either (ایک دوسرے پر) of There two forces may be called action. Then The other force will be re-action:-

Friction and Normal Reaction:-

"Suppose two bodies 'A' and 'B' are Contact at point 'C'



If The body 'A' is exerts a force $\vec{G}$ on The body 'B' Then The Body 'B' exerts The force $-\vec{G}$ on The body 'A'

Then either of forces may be called action and the other will be reaction. The force $\vec{G}$ generaly makes a certain angle to the Common normal at the pt of Contact "C" Suppose $\vec{F}$ and $\vec{R}$ are the resolved parts of '$\vec{G}$' in the tangent Plane and along the Common Normal at 'c' respectively.

The $\vec{F}$ is called friction force and $\vec{R}$ is called Normal reaction.

Force of friction or simply friction

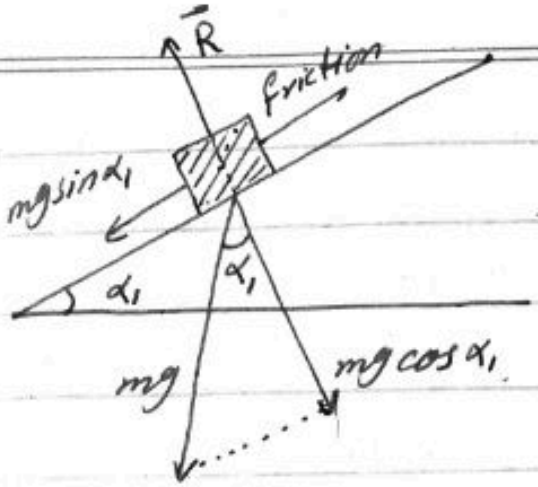
Types of Friction:-

There are three types of friction.

(i)

Static Friction:-

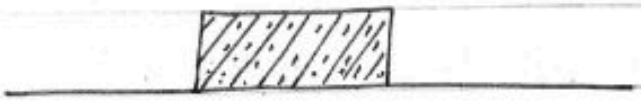
When two bodies are in Contact with each other but neither body is on the point of sliding along the Then friction is this case is called



static friction

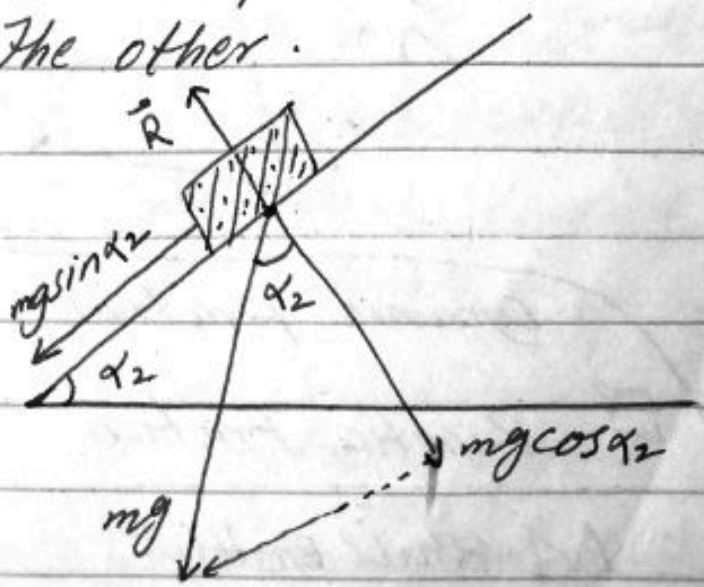
OR

Non-Limiting friction:-



(ii) Limiting Friction:-

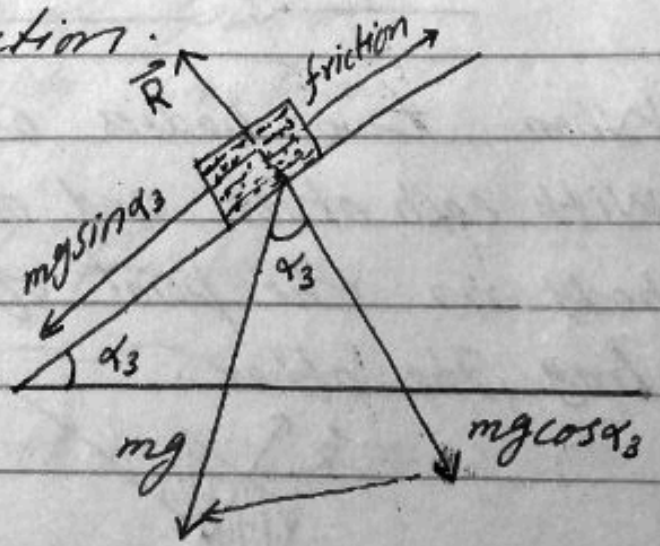
"When two bodies are in Contact With each other and one of the body in the point of sliding along the other.



Then the friction in this case is called limiting friction. In such a case the equilibrium of the body is said to be limiting equilibrium.

(iii) Sliding Friction:-

When two bodies are in Contact with each other and are one body sliding along the other, Then the friction in this case is called sliding friction.



OR

Dynamic Friction

OR

Kinetic Friction

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Remarks :-

"The magnitude of the dynamic friction" is slightly less than the limiting friction"

Law of friction :-

Friction :-

"A force which opposes the motion of a body is called friction or force of friction."

(i) The direction of friction is opposite to the direction in which the body moves or tend to move.

(ii) The magnitude of friction is upto a certain extent (to) equal to the force tending to produe to the motion :-

(iii) Only a certain amount of firction can be called in to play and in each particular it can-not

be exceed (حد سے گزر جانا، تجاوز کرنا) a certain limit

(iv) If 'F' is limiting friction 'R' is the normal reaction:-

Then $\frac{F}{R} = \mu$

Where 'μ' is Constant Know is Co-efficient of friction.

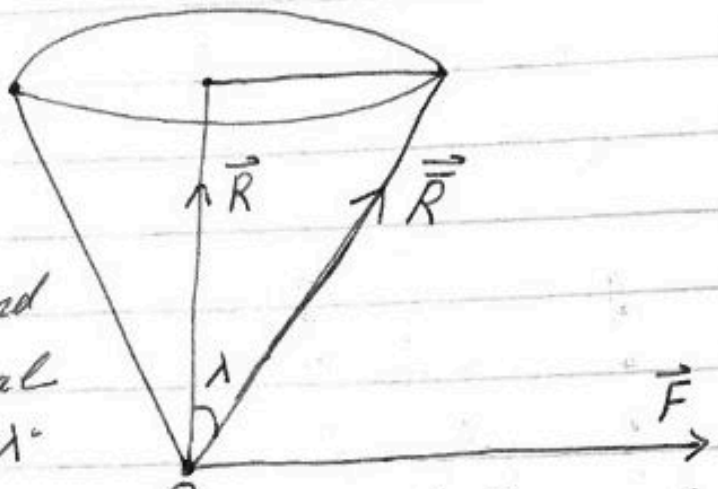
(v) The amount of friction is Independent of the areas and shape of the surfaces in Contact Provided the normal pressure remains Constant / the same.

(vi) When the motion take place the friction still opposes the motion. It is Independent the velocity and slightly less than the limiting friction.

عقل مند وہ ہے، جو اپنی زبان
کو دوسروں کی مذمت سے بچائے رکھے

Cone of Friction:-

We draw a Cone with vertex at 'O' The pt of Contact of two bodies and semivertical angle 'λ'



Then This Cone is called Cone of friction. Since $\vec{R}$ Can makes any angle with $\vec{R}$ with magnitude upto 'λ' Therefore $\vec{R}$ must be on or with in The Cone.

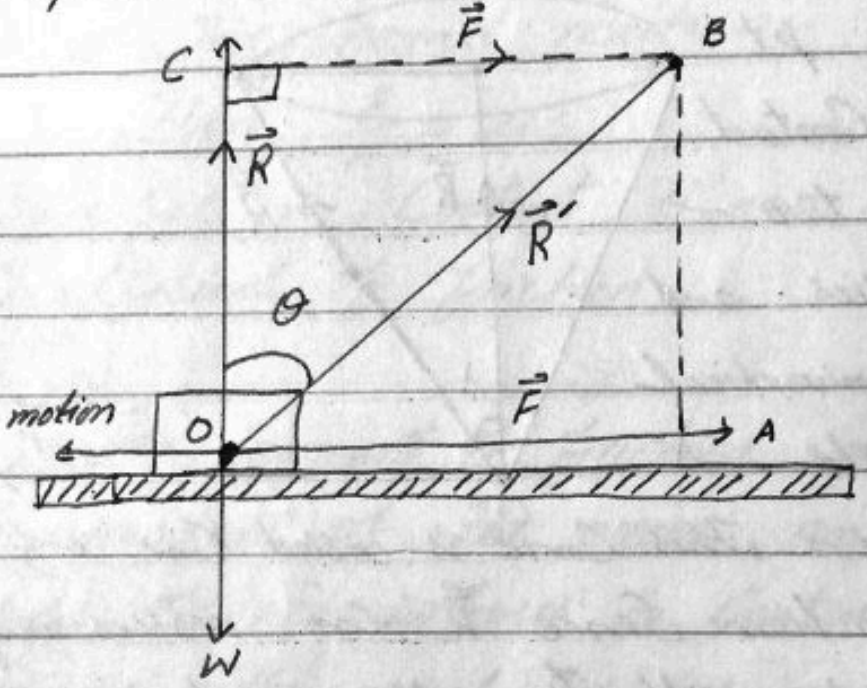
Friction:- The friction plays an imp role in every day of life.

When a person walkis he pushes the ground backward with his feet to cause a forward reaction. ∴

∴ If there were no friction his feet would slip and he would not able to walk.

Angle of Friction:-

Consider a body of weight 'w' is placed on a rough surface.



Let $\vec{F}$ be the force of friction and $\vec{R}$ be the normal pressure/friction Let $\vec{R}'$ be the resultant force of the Concurrent forces $\vec{F}$ and $\vec{R}$. Complete the parallelogram "OABC" Clearly OCB is a right angleal triangle 'Then'

$$\tan\theta = \frac{F}{R} \quad \text{Put } \frac{F}{R} = \mu$$

$$\tan\theta = \mu \Rightarrow \theta = \tan^{-1}(\mu)$$

θ is called angle of friction and is generally denoted by 'λ'

i.e $\theta = \lambda = \tan^{-1}(\mu)$

The angle between the normal pressure $\vec{R}$ and $\vec{R'}$ (The resultant of $\vec{R}$ and $\vec{F}$) is called angle of friction.

★ Smooth Contact :-

"In case of the friction $\vec{F}$ b/w two bodies in Contact with each other is zero, The Contact is said to be smooth."

Smooth Body :-

If The friction force b/w two bodies (X) "If a body can have only a smooth Contact with every other body, It is said to be a smooth body"

Rough Bodies :-

If The friction b/w two bodies in Contact is not zero. Bodies are said to be rough

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(1) :- Find the Condition for Limiting equilibrium of a body under its own weight is a rough inclined Plane

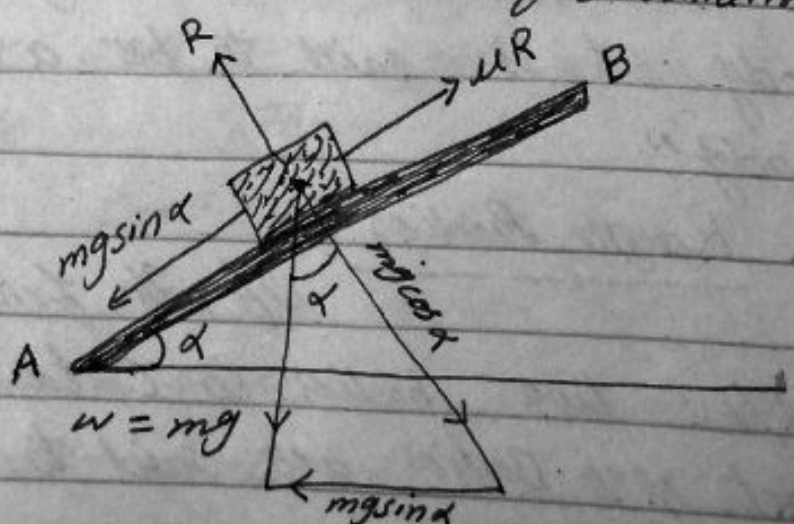
OR

If a Paricle is in limiting equilibrium on a ~~particle~~ inclined plane under its own weight

Then the Inclination of the ~~force~~ Plane equals the magnitude of the angle of friction :-

Proof

" Let a Particle of mass 'm' be in limiting equilibrium on an inclined plane AB of Inclination 'α',



∴ Let 'μ' be the Co-efficient of friction and 'λ' be the angle of friction Let R be the normal reaction:-

Resolving the forces along and perpendicular to the inclined plane we get

$$mg\sin\alpha = \mu R \quad \text{——— (1)}$$

$$mg\cos\alpha = R \quad \text{——— (2)}$$

Dividing (1) by (2)

$$\frac{mg\sin\alpha}{mg\cos\alpha} = \frac{\mu R}{R}$$

$$\Rightarrow \tan\alpha = \mu$$

$$\Rightarrow \tan\alpha = \tan\lambda$$

$$\because \mu = \tan\lambda$$

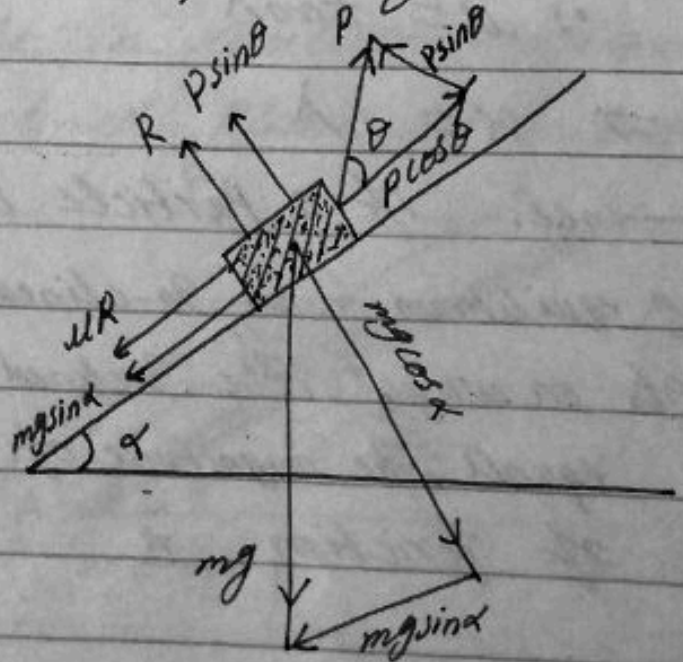
$$\Rightarrow \alpha = \lambda$$

∴ Hence:- if a particle be in limiting equilibrum on an In-clined plane under its on weight, the Inclined of the plane equals the magnitude of the angle of friction. ∎

i-e/ Angle of inclination = Angle of friction
Clearly if a body is in equilibrium on an inclined plane under its weight then this is possible only when the angle of inclination is equal to the angle of friction.

(2) :- Find the least force required to drag up a particle on a rough inclined plane.

Sol:
Suppose a force 'P' making an angle 'θ' with the plane, can just move a particle of mass 'm' up an inclined plane of inclination 'α'.



Let the particle is in limiting equilibrium. Let 'u' be the co-efficient of friction and 'R' the normal reaction. Let 'λ' be the angle of friction.

Resolving the forces along and perpendicular to plane, we get,

$$P\cos\theta = uR + mg\sin\alpha \quad \text{—— (1)}$$

$$P\sin\theta + R = mg\cos\alpha \quad \text{—— (2)}$$

From (2) $R = mg\cos\alpha - P\sin\theta$

Put in eq (1)

$$P\cos\theta = u(mg\cos\alpha - P\sin\theta) + mg\sin\alpha$$

$$P\cos\theta = umg\cos\alpha - uP\sin\theta + mg\sin\alpha$$

$$P\cos\theta + uP\sin\theta = umg\cos\alpha + mg\sin\alpha$$

$$P(\cos\theta + u\sin\theta) = mg(u\cos\alpha + \sin\alpha)$$

$$P(\cos\theta + u\sin\theta) = mg(u\cos\alpha + \sin\alpha)$$

$$P = mg \cdot \frac{u\cos\alpha + \sin\alpha}{\cos\theta + u\sin\theta}$$

$$P = mg \cdot \frac{\tan\lambda\cos\alpha + \sin\alpha}{\cos\theta + \tan\lambda\sin\theta}$$

$$\because u = \tan\lambda$$

$$P = mg \frac{\frac{\sin\lambda}{\cos\lambda}\cdot\cos\alpha + \sin\alpha}{\cos\theta + \frac{\sin\lambda}{\cos\lambda}\sin\theta}$$

$$P = mg \frac{\sin\lambda\cos\alpha + \cos\lambda\sin\alpha}{\cos\lambda\cos\theta + \sin\lambda\sin\theta}$$

$$P = mg \frac{\sin(\alpha+\lambda)}{\cos(\theta-\lambda)} \quad \text{---} \quad (3)$$

Now 'P' is Least of L.H.S of above eq is least,

or $\cos(\theta-\lambda)$ is maximum

or $\cos(\theta-\lambda) = 1 \quad \because \cos(0) = 1$

or $\cos(\theta-\lambda) = \cos(0)$

or $\theta - \lambda = 0$

$\theta = \lambda$

So, eq (3) be Comes

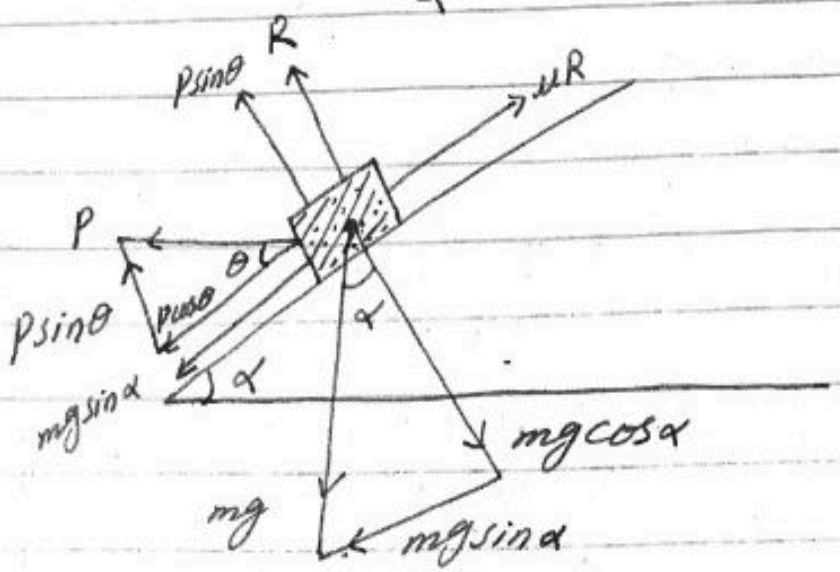
$$P_{least} = mg\sin(\alpha+\lambda)$$

(3) :- Find The Least force - required to drag the particle down The plane?

Sol:

Suppose a force P making an angle 'θ' with The plane, Can just move a particle of mass 'm" down an inclined plane of in-clination

"α"



Suppose the particle is in limiting friction, Let μ, R, λ be the Co-efficient of friction normal reaction and angle of friction respectively.

Re-solving the forces onlong and perpendicular to the inclination // inclined plane, we get

$$\mu R = mg\sin\alpha + P\cos\theta \quad \text{—— (1)}$$

$$P\sin\theta + R = mg\cos\alpha \quad \text{—— (2)}$$

from (2)

$$R = mg\cos\alpha - P\cos\theta$$

Put in (1)

$$\mu(mg\cos\alpha - P\cos\theta) = mg\sin\alpha + P\cos\theta$$

$$\mu mg\cos\alpha - \mu P\sin\theta = mg\sin\alpha + P\cos\theta$$

$$\Rightarrow \mu P\sin\theta + P\cos\theta = \mu mg\cos\alpha - mg\sin\alpha$$

$$\Rightarrow P(\mu\sin\theta + \cos\theta) = mg(\mu\cos\alpha - \sin\alpha)$$

$$\Rightarrow P = mg\frac{\mu\cos\alpha - \sin\alpha}{\mu\sin\theta + \cos\theta}$$

$$\Rightarrow P = mg\frac{\tan\lambda\cos\alpha - \sin\alpha}{\tan\lambda\sin\theta + \cos\theta}$$

$$\Rightarrow P = mg\frac{\frac{\sin\lambda}{\cos\lambda}\cos\alpha - \sin\alpha}{\frac{\sin\lambda}{\cos\lambda}\sin\theta + \cos\theta}$$

$$\Rightarrow P = mg\frac{\sin\lambda\cos\alpha - \sin\alpha\cos\lambda}{\sin\lambda\sin\theta + \cos\theta\cos\lambda}$$

$$\Rightarrow P = mg\frac{\sin(\lambda-\alpha)}{\cos(\lambda-\theta)} \quad \text{---------- (3)}$$

Now 'P' is least, if R.H.S is minimum

or $\cos(\lambda-\theta)$ is maximum

or $\cos(\lambda-\theta) = 1$

$$\cos(\lambda-\theta) = \cos(0)$$

$$\because \cos(0) = 1$$

$$\lambda - \theta = 0$$

$$\lambda = \theta$$

So eq (3) become, $P_{least} = mg(\sin(\lambda-\alpha)$

$$P_{least} = mg\sin(\lambda-\alpha)$$

Q Find the least force to support a heavy particle on an inclined plane??

Sol:

Suppose a force 'P' making an angle 'θ' with the plane. Can just support the particle on an inclined plane of inclination $\alpha > \lambda$ where 'λ' is angle of friction 'ε'. Resolving the forces along and perpendicular to the plane, we get

$$P\cos\theta + \mu R = mg\sin\alpha \quad \text{——— (1)}$$

$$P\sin\theta + R = mg\cos\alpha \quad \text{——— (2)}$$

From (2)

$$R = mg\cos\alpha - P\sin\theta$$

Put in eq (1)

$$P\cos\theta + \mu(mg\cos\alpha - P\sin\theta) = mg\sin\alpha$$

$$P\cos\theta + \mu mg\cos\alpha - \mu P\sin\theta = mg\sin\alpha$$

$$\Rightarrow P\cos\theta + \mu mg\cos\alpha - \mu P\sin\theta = mg\sin\alpha$$

$$\Rightarrow P\cos\theta - \mu P\sin\theta = mg\sin\alpha - \mu mg\cos\alpha$$

$$\Rightarrow P(\cos\theta - \mu\sin\theta) = mg(\sin\alpha - \mu\cos\alpha)$$

$$\Rightarrow P = mg\,\frac{\sin\alpha - \mu\cos\alpha}{\cos\theta - \mu\sin\theta}$$

$$\Rightarrow P = \frac{mg\sin\alpha - \tan\lambda\cos\alpha}{\cos\theta - \tan\lambda\sin\theta}$$

$$\Rightarrow P = mg \cdot \frac{\sin\alpha - \frac{\sin\lambda}{\cos\lambda}\cos\alpha}{\cos\theta - \frac{\sin\lambda}{\cos\lambda}\sin\theta}$$

$$\Rightarrow P = mg \cdot \frac{\sin\alpha\cdot\cos\lambda - \sin\lambda\cos\alpha}{\cos\theta\cos\lambda - \sin\theta\sin\lambda}$$

$$\Rightarrow P = mg\,\frac{\sin(\alpha-\lambda)}{\cos(\theta+\lambda)} \quad \text{---------- (3)}$$

Now, P is least if R.H.S is minimum,

or $\cos(\theta+\lambda)$ is maximum

or $\cos(\theta+\lambda) = 1$

or $\cos(\theta+\lambda) = \cos(0)$

$$\theta+\lambda = 0$$

$$\theta = -\lambda$$

So, eq (3) becomes

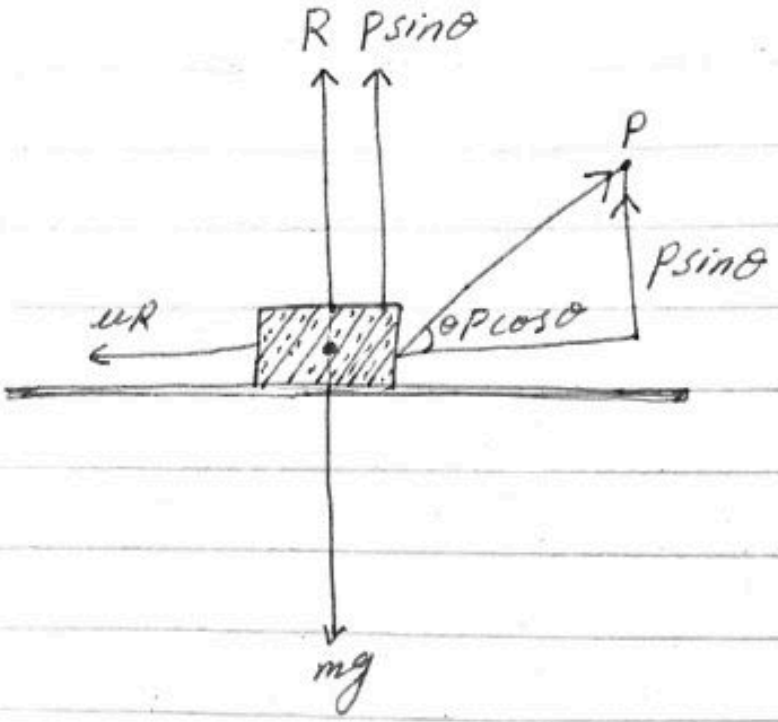
$$P_{least} = mg\sin(\alpha-\lambda)$$

مکافات عمل کی چکی چلتی آہستہ ہے
ہاں۔ مگر پیشتی بہت باریک ہے

(5) :- Find the least force to set a particale in to motion on an horizontal plane.

Sol:

Suppose a force 'P' making an angle 'θ' with the plane, can just move a particle of mass 'm' on a horizontal plane.



Let μ be the co-efficient of friction
Let R and λ be the normal reaction and angle of friction resp.
Resolving the forces along and Perpendicular to the plane, we get

$P\cos\theta = \mu R$ ——— (1)

$mg = P\sin\theta + R$ — (2)

from eq (2)

$R = mg - P\sin\theta$

put in eq (1)

$P\cos\theta = \mu(mg - P\sin\theta)$

$\Rightarrow P\cos\theta = \mu mg - \mu P\sin\theta$

$\Rightarrow P\cos\theta + \mu P\sin\theta = \mu mg$

$\Rightarrow P(\cos\theta + \mu\sin\theta) = \mu mg$

$$\Rightarrow P = mg \cdot \frac{\mu}{\cos\theta + \mu\sin\theta}$$

$$\Rightarrow P = mg \cdot \frac{\tan\lambda}{\cos\theta + \tan\lambda\sin\theta}$$

$$\Rightarrow P = mg \cdot \frac{\frac{\sin\lambda}{\cos\lambda}}{\cos\theta + \frac{\sin\lambda}{\cos\lambda}\sin\theta}$$

$$\Rightarrow P = mg \frac{\sin\lambda}{\cos\theta\cos\lambda + \sin\theta\cdot\sin\lambda}$$

$$\Rightarrow P = mg \frac{\sin\lambda}{\cos(\theta - \lambda)} \quad \text{——— (3)}$$

Now, 'P' is least if R.H.S is minimum

or $\cos(\theta - \lambda)$ is maximum

or $\cos(\theta-\lambda) = 1$

or $\cos(\theta-\lambda) = \cos(0)$

or $(\theta-\lambda) = 0$

$\theta = \lambda$

So, eq (3) becomes

$$P_{least} = mg\sin\lambda$$

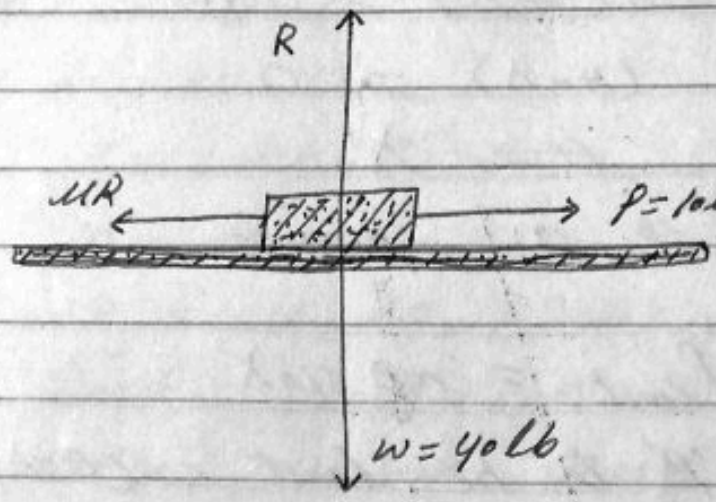
for 'P' to be least $\cos(\theta-\lambda)$ must be maximum, and the maximum value of $\cos(\theta-\lambda) = 1$

(6) :-

A body of weight 40 lb is resting on a rough horizental plane and can just be moved by a force of 10 lb weight. acting horizentally Find the Coefficient of friction.

Sol :- Consider a body of weight $w = 40$ lb is placed on a rough horizental surface. Let μ is Co-efficient of friction. when the particle is just on the point of moving with the help of a force

of 10 lb acting horizontally



The various forces acting on the system are show in fig.

Re-solving the forces :-

$$R = W \quad \text{—— (1)}$$

$$P = \mu R \quad \text{—— (2)}$$

$\because W = 40lb, \quad P = 10\,lb$

Dividing (2) by (1)

$$\frac{P}{R} = \frac{\mu R}{W} \Rightarrow \underline{OR}$$

$$\frac{\mu R}{R} = \frac{10lb}{40lb} \Rightarrow \mu = \frac{1}{4} = 0.25$$

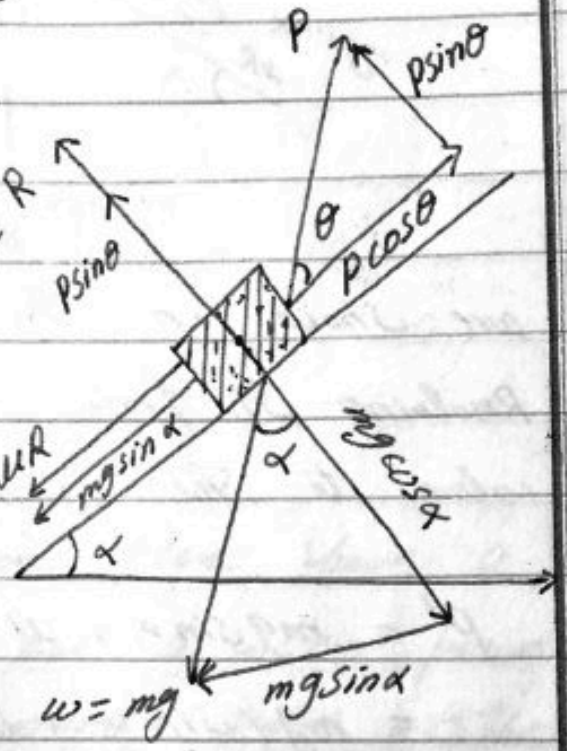
$$\mu = 0.25$$

(7) :- The Least force which will move a weight up an inclined plane

is of magnitude 'P' show that the least force acting parallel to the plane, which will move the weight is $P\sqrt{1+\mu^2}$ where μ is the co-efficient of friction.

Sol^{n} :-

Consider a body of weight $w=mg$ is placed on an inclination with the horizontal ~~force which~~ is 'α'

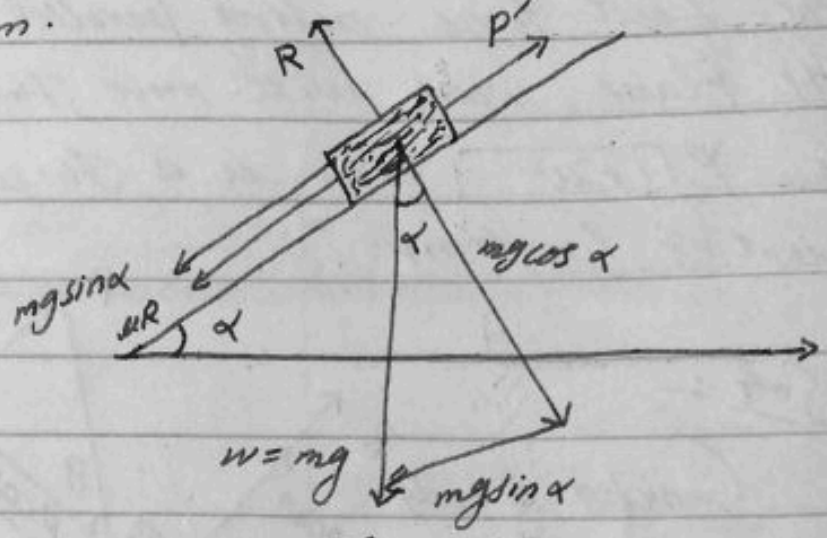


Let 'P' be the least force which drags the body up on the inclined plane then.

$$P = mg\sin(\lambda+\alpha) \quad \text{---} \quad (1)$$

Let P' be the least force acting parallel to the inclined plane and drag the weight 'w' on the same inclined plane.

The various forces acting on the system.



are show in the fig.

Resolving the forces along and perpendicular to the plane, we get

$$P' = mg\sin\alpha + \mu mg\cos\alpha$$
$$= mg(\sin\alpha + \mu\cos\alpha)$$
$$= mg(\sin\alpha + \tan\lambda\cos\alpha)$$
$$= mg\left(\sin\alpha + \frac{\sin\lambda}{\cos\lambda}\cos\alpha\right)$$

$$P' = mg\,\frac{\sin(\alpha+\lambda)}{\cos\lambda} \quad \text{——— (3)}$$

We know that the least force 'P' making angle 'θ' with plane, to move a particle of mass 'm' up on the inclined plane of inclination 'α' is given by

$$P = mg\sin(\alpha+\lambda)$$

So (3) becomes

$$P' = \frac{P}{\cos\lambda}$$

$$P' = P\sec\lambda$$

$$P' = P\sqrt{\sec^2\lambda}$$

$$P' = P\sqrt{1+\tan^2\lambda}$$

$$P' = P\sqrt{1+\mu^2}$$

Proved

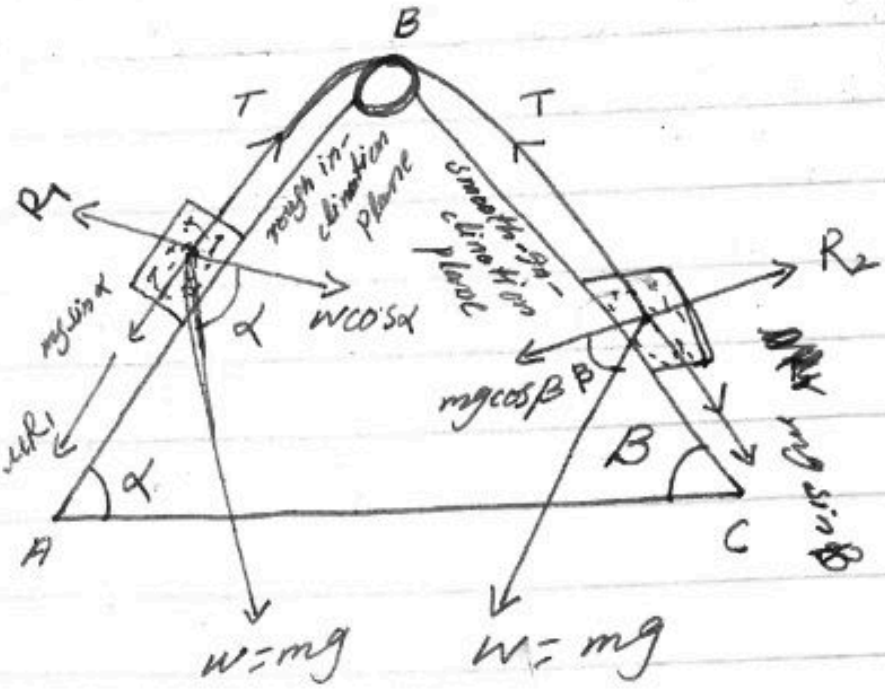
☆

(1) :- Two inclined plane have a common vertex, and a string passing over a small smooth pully at the vertex. supports two equal weight If one of the plene be rough and the other smooth. find the relation b/w the inclination of the planes when the weight on the smooth plane is on the point of moving down.

Sol. :-

Consider two weight w, w' placed on two inclined planes AB

and BC as show in fig



Let w be the weight of either particle. Let α and β be the inclinations of rought and smooth plane resp.

Let 'u' be the co-efficient of friction of rough plane.

Let R_1 & R_2 be the normal reactions of rought and smooth Planes resp. Let 'T' be the tension in the string For the particle on the roughPlane.

For the particle on rough plane.

Resolving the forces along and perpendicular to the rough plane we get.

$$T = \mu R_1 + W\sin\alpha \quad \text{——(1)}$$

$$R_1 = W\cos\alpha \quad \text{——(2)}$$

Put R_1 in (1)

$$T = \mu(W\cos\alpha) + W\sin\alpha$$

$$T = \mu W\cos\alpha + W\sin\alpha \quad \text{——(3)}$$

For a particle on smooth plane= Resolving the forces along and perpendicular to the smooth plane, we get,

$$T = W\sin\beta \quad \text{——(4)}$$

$$R_2 = W\cos\beta \quad \text{——(5)}$$

Comparing (3) & (4)

$$W\sin\beta = \mu W\cos\alpha + W\sin\alpha$$

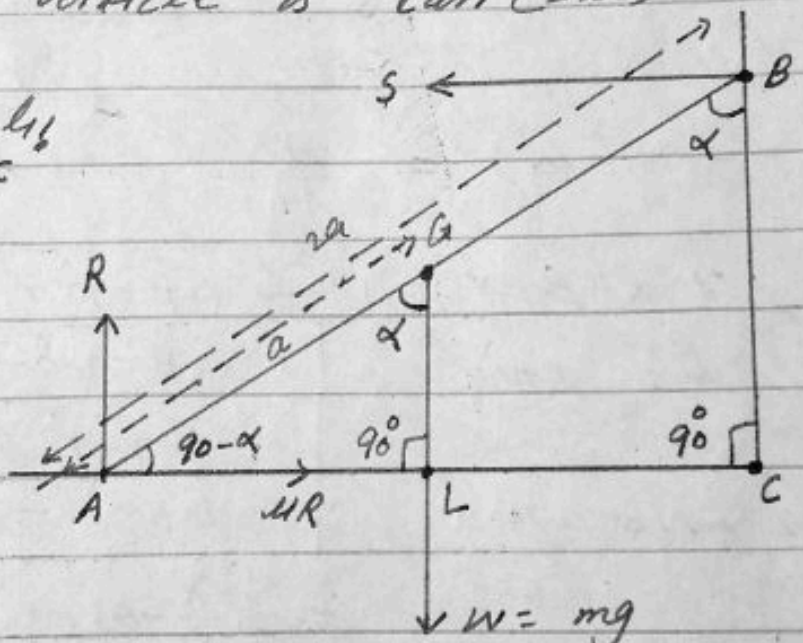
$$W\sin\beta = W(\mu\cos\alpha + \sin\alpha)$$

$$\sin\beta = \mu\cos\alpha + \sin\alpha$$

which is the relation b/w α and β.

Q:- A uniform ladder rests in limiting equilibrium with one end on a rough floors, whose Co-efficient of friction is 'u' and with the other against a smooth vertical wall. Show that its in-clination to verticle is $\tan^{-1}(2u)$

Sol:



Let 'α' be the inclination of the ladder to the vertical.
Let '2a' be the lenght of and 'W' be the weight of the ladder.
Let R and S be the normal reaction of floor and wall resp.
Re-solving the forces horizantally and vertically. we get-

$S = \mu R$ — (1)

$W = R$ — (2)

$S = \mu W$ — (3)

Now, Taking the moments about 'A'

★ The turning effect of a force is called moment.

$W \cdot AN = S \cdot BC$ — (4)

$\cos(90^\circ - \alpha) = \frac{AN}{AM}$

$\because \alpha + \beta + \gamma = 180 \Rightarrow \alpha + \beta + 90^\circ = 180^\circ$

$\sin\alpha = \frac{AN}{a}$

$\Rightarrow AN = a\sin\alpha$

$\alpha + \beta = 180^\circ - 90^\circ$

$\alpha + \beta = 90^\circ$

$\beta = 90^\circ - \alpha$

$\because \cos(\frac{\pi}{2} - \theta) = \sin\theta$

Also, $\sin(90^\circ - \alpha) = \frac{BC}{AB}$

$\Rightarrow \cos\alpha = \frac{BC}{2a}$

$\Rightarrow BC = 2a\cos\alpha$

Put $BC = 2a\cos\alpha$ in eq (4) & $AN = a\sin\alpha$

$\Rightarrow W \cdot a\sin\alpha = S \cdot 2a\cos\alpha$

$\Rightarrow W\sin\alpha = 2S\cos\alpha$

$\Rightarrow W\sin\alpha = 2\mu W\cos\alpha$

by (3)

$\Rightarrow \sin\alpha = 2\mu\cos\alpha$

$\Rightarrow 2\mu = \frac{\sin\alpha}{\cos\alpha}$

$\Rightarrow 2\mu = \tan\alpha$

$\Rightarrow \alpha = \tan^{-1}(2\mu)$

Proved

Q:-

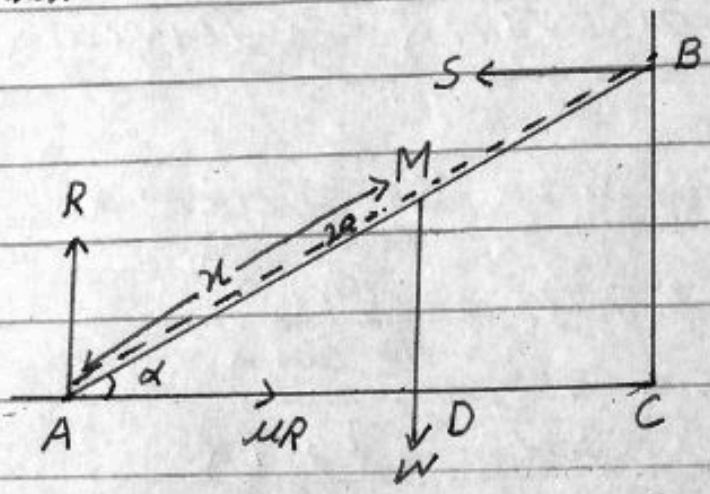
"A Light ladder is supported on a rough floor and leans against a smooth wall. How far up the ladder can a man climb without slipping taking place?

Sol. Consider a Light Ladder $\overline{AB}$ of Lenght 'l' rests with one end on rough floor and other leans a smooth vertical wall, with makes an angle 'α'

OR

Let 'x' be the distance of

The man 'M' when the ladder slip.



Let 'W'' be the weight of man. Let Lenght of the ladder AB is 2a. Let 'μ' be the co-efficient of friction of floor. Let R & S be the normal reaction of floor and the wall resp.
resolving the forces horizentally and vertically we get.

$$R = W \quad \text{---(1)}$$

$$S = \mu R \quad \text{---(2)}$$

$$S = \mu W \quad \text{---(3)}$$

So, Taking moment of the forces about 'A'

$$W \cdot AD = S \cdot BC \quad \text{---(4)}$$

In right $\triangle ADM$

$$\cos\alpha = \frac{AD}{AM} \Rightarrow \cos\alpha = \frac{AD}{x}$$

$$\Rightarrow AD = x\cos\alpha$$

In right ΔABC

$$\sin\alpha = \frac{BC}{AB} \Rightarrow \sin\alpha = \frac{BC}{2a}$$

$$\Rightarrow BC = 2a\sin\alpha$$

Put in (4)

$$Wx\cos\alpha = S \cdot 2a\sin\alpha$$

$$\because S = \mu W$$

$$Wx\cos\alpha = \mu W 2a\sin\alpha$$

$$x\cos\alpha = 2a\mu\sin\alpha$$

$$x = 2a\mu\frac{\sin\alpha}{\cos\alpha}$$

$$x = 2a\mu\tan\alpha$$

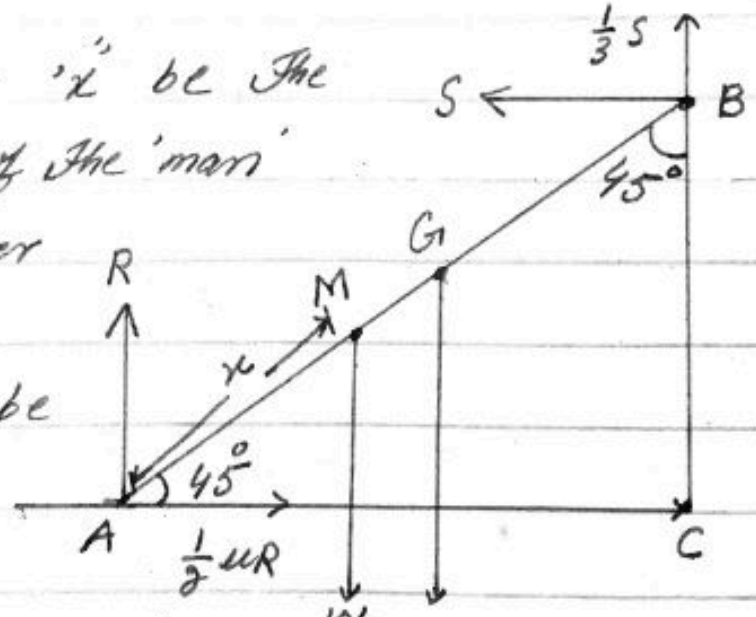
required result.

Q:- A uniform ladder of lenght 70 ft rests against a vertical wall with it makes an angle '45°'. The Co-efficient of friction b/w the ladder and wall and the ground resp. being $\frac{1}{3}$ and $\frac{1}{2}$. If a man whose weight is one half of the ladder ascends the ladder where will he be when the ladder slips.?

Sol:

Consider 'x' be the distance of the 'man' when ladder slips.

Let 'W' be the weight of ladder $\bar{AB}$. The weight of man will be $W/2$. Let R and S be the normal reaction of floor and wall resp.



Resolving the forces horizent and vertically. we get

$$\frac{1}{2}R = S$$

OR $R = 2S$ —— (1)

$$\frac{W}{2} + W = R + \frac{1}{3}S$$

OR $\frac{3}{2}W = R + \frac{1}{3}S$ —— (2)

Now Taking moments about 'A'

$$\frac{W}{2} \cdot AL + W \cdot AN = S \cdot BC + \frac{1}{3}S \cdot AC - (3)$$

In $\triangle ALM$

$$\cos 45^\circ = \frac{AL}{AM} \Rightarrow \frac{1}{\sqrt{2}} = \frac{AL}{x} \Rightarrow AL = \frac{x}{\sqrt{2}}$$

In $\triangle ANG$

$$\cos 45^\circ = \frac{AN}{AG} = \frac{1}{\sqrt{2}} = \frac{AN}{35} \Rightarrow AN = \frac{35}{\sqrt{2}}$$

In $\triangle ABC$

$$\sin 45^\circ = \frac{BC}{AB} \Rightarrow \frac{1}{\sqrt{2}} \Rightarrow \frac{BC}{70} \Rightarrow BC = \frac{70}{\sqrt{2}}$$

In $\triangle ABC$

$$\cos 45^\circ = \frac{AC}{AB} \Rightarrow \frac{1}{\sqrt{2}} = \frac{AC}{70} \Rightarrow AC = \frac{70}{\sqrt{2}}$$

Put in (3)

$$\frac{W}{2} \cdot \frac{x}{\sqrt{2}} + W\frac{35}{\sqrt{2}} = S \cdot \frac{70}{\sqrt{2}} + \frac{1}{3}S \cdot \frac{70}{\sqrt{2}}$$

$$\Rightarrow \frac{xW}{2\sqrt{2}} + \frac{35W}{\sqrt{2}} = \frac{70S}{\sqrt{2}} + \frac{70S}{3\sqrt{2}}$$

$$\Rightarrow W\left(\frac{x}{2\sqrt{2}} + \frac{35}{\sqrt{2}}\right) = S\left(\frac{70}{\sqrt{2}} + \frac{70}{3\sqrt{2}}\right)$$

$$\Rightarrow \frac{W}{12}\left(\frac{x+70}{2}\right) = \frac{S}{12}\left(\frac{210+70}{3}\right)$$

$$\Rightarrow W\left(\frac{x+70}{2}\right) = S\left(\frac{280}{3}\right) \quad \text{—(4)}$$

Eliminating R from (1) & (2)

$$\frac{3W}{2} = 2S + \frac{1}{3}S$$

$$\Rightarrow \frac{3W}{2} = \frac{7S}{3} \Rightarrow S = \frac{3W}{2} \times \frac{3}{7}$$

$$\Rightarrow S = \frac{9W}{14} \quad \text{put in (4)}$$

$$W\left(\frac{x+70}{2}\right) = \frac{9W}{14} \cdot \frac{280}{3}$$

$$\Rightarrow \frac{x+70}{2} = 60$$

$$\Rightarrow x+70 = 120 \Rightarrow x = 120-70$$

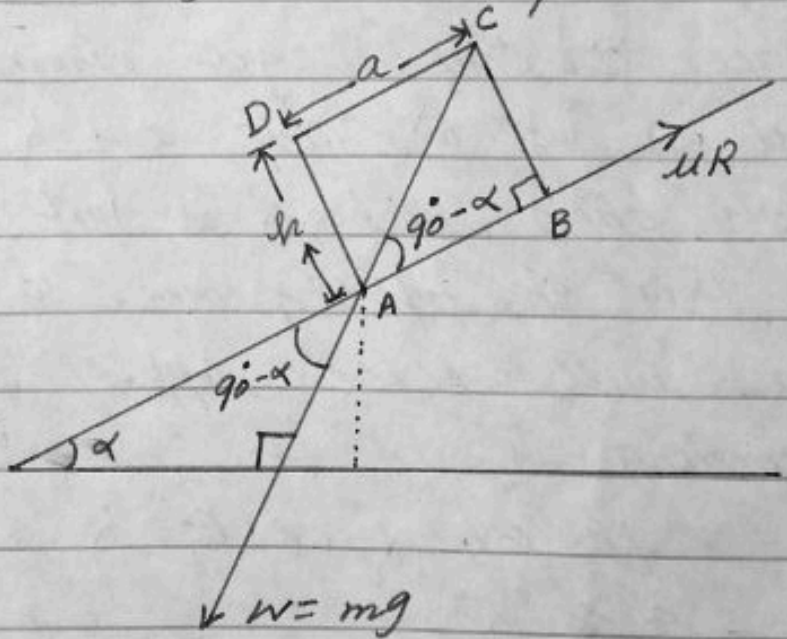
$$x = 50 \text{ ft} \quad \underline{\text{Ans}}$$

Q:- A uniform rectangular block of height, "h" whose base is a square of side 'a' rest on a rough inclined plane. The plane is gradually titled about a line parallel to two edges of the base. Show that the block will slide or topple over according

as $a \geq uh$, u is Co-efficient of friction.

Sol!:-

A uniform rectangular block of height 'h' whose base is a square shape of side 'a' rests on a rough inclined plane.



Let 'α' be the inclination of the inclined plane to the ~~point of toppling~~ horizental when the block is just on the point of toppling over. From right angled triangle ABC

$$\frac{AB}{BC} = \cot(90^\circ - \alpha)$$

$$\frac{a}{h} = \tan\alpha \quad \text{---------- (1)}$$

Now θ be the inclination of the inclined plane to horizental when block is about to slide.

Then $\theta = \lambda$ (condition of limiting equilibrium)

$$\tan\theta = \tan\lambda$$

Put $\tan\lambda = \mu$

$$\tan\theta = \mu \quad \text{---} \quad (2)$$

Since 'θ' be the angle when the block is just to slide and 'α' the angle when the block is just on the point of toppling over, so the block will slide a topple over according as

if $\theta < \alpha$ or $\theta > \alpha$

$\theta \gtrless \alpha$

$\tan\theta \gtrless \tan\alpha$

by (i) & (2)

$\mu \gtrless \frac{a}{h}$

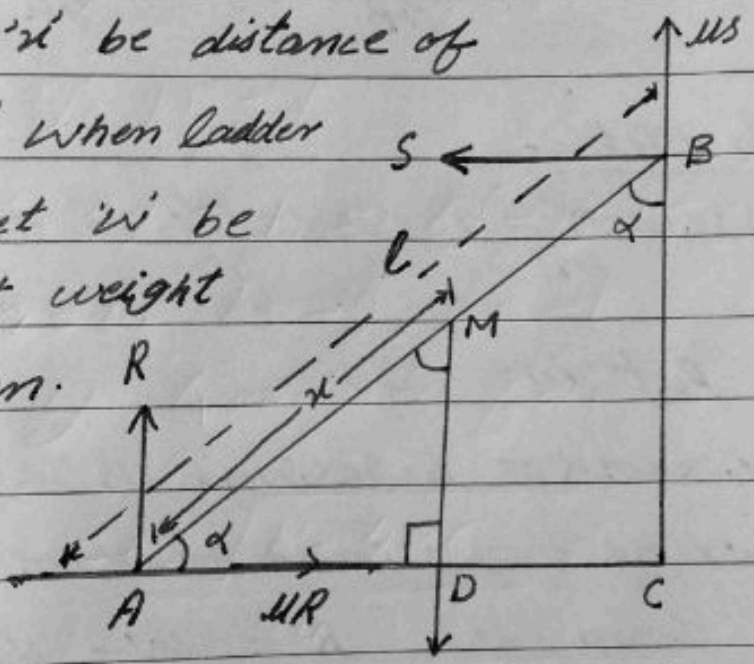
$\times h$ both side

$h\mu \gtrless a$

or $a \gtrless \mu h$

Q:- A light ladder is supported on a rough floor and leans against s equally rough wall. How far up the ladder can a man climb without slipping taking place? (Here 'l' is the length of the ladder and 'α' its inclination with the vertical)

Sol:- Let 'x' be distance of man 'M' when ladder slips. Let 'w' be the ~~lenght~~ weight of the man.



Let '2a' be the length of Ladder AB. Let R & S be the normal reaction of floor and wall respectively. Resolving the forces horizentally and vertically we get

$$R + \mu S = W \quad \text{---} \quad (1)$$

$$S = \mu R \quad \text{---} \quad (2)$$

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Taking moments about 'A'

$$W \cdot AD = S \cdot BC + \mu S \cdot AC \quad \text{——} (3)$$

In $\triangle ADM$

$$\cos\alpha = \frac{AD}{AM} \Rightarrow \cos\alpha = \frac{AD}{x} \Rightarrow AD = x\cos\alpha$$

In $\triangle ABC$

$$\sin\alpha = \frac{BC}{AB} \Rightarrow \sin\alpha = \frac{BC}{2a} \Rightarrow BC = 2a\sin\alpha$$

In $\triangle ABC$

$$\cos\alpha = \frac{AC}{AB} \Rightarrow \cos\alpha = \frac{AC}{2a} \Rightarrow AC = 2a\cos\alpha$$

Put in (3)

$$W \cdot x\cos\alpha = S \cdot 2a\sin\alpha + \mu S\, 2a\cos\alpha$$

$$\Rightarrow Wx\cos\alpha = 2aS\sin\alpha + 2a\mu S\cos\alpha \quad \text{——} (4)$$

from (1) $R = W - \mu S$

Put in eq (2)

$$S = \mu(W - \mu S)$$

$$\Rightarrow S = \mu W - \mu^2 S$$

$$\Rightarrow S + \mu^2 S = \mu W$$

$$\Rightarrow S(1 + \mu^2) = \mu W$$

$$\Rightarrow S = \frac{\mu W}{1 + \mu^2}$$

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Put in eq (4)

$$W x \cos\alpha = 2a\left(\frac{\mu W}{1+\mu^2}\right)\sin\alpha + 2a\mu\left(\frac{\mu W}{1+\mu^2}\right)\cos\alpha$$

$$\Rightarrow x\cos\alpha = \frac{2a\mu}{1+\mu^2}\sin\alpha + \frac{2a\mu^2}{1+\mu^2}\cos\alpha$$

$$\Rightarrow x\cos\alpha = \frac{2a\mu}{1+\mu^2}(\sin\alpha + \mu\cos\alpha)$$

$$\Rightarrow x = \frac{2a\mu}{1+\mu^2}(\tan\alpha + \mu)$$

ANS

Q:- A rod 4ft long rest on a rought floor against the

Put in eq (4)

$$W\times\cos\alpha = 2a\left(\frac{\mu W}{1+\mu^2}\right)\sin\alpha + 2a\mu\left(\frac{\mu W}{1+\mu^2}\right)\cos\alpha$$

$$\Rightarrow x\cos\alpha = \frac{2a\mu}{1+\mu^2}\sin\alpha + \frac{2a\mu^2}{1+\mu^2}\cos\alpha$$

$$\Rightarrow x\cos\alpha = \frac{2a\mu}{1+\mu^2}(\sin\alpha + \mu\cos\alpha)$$

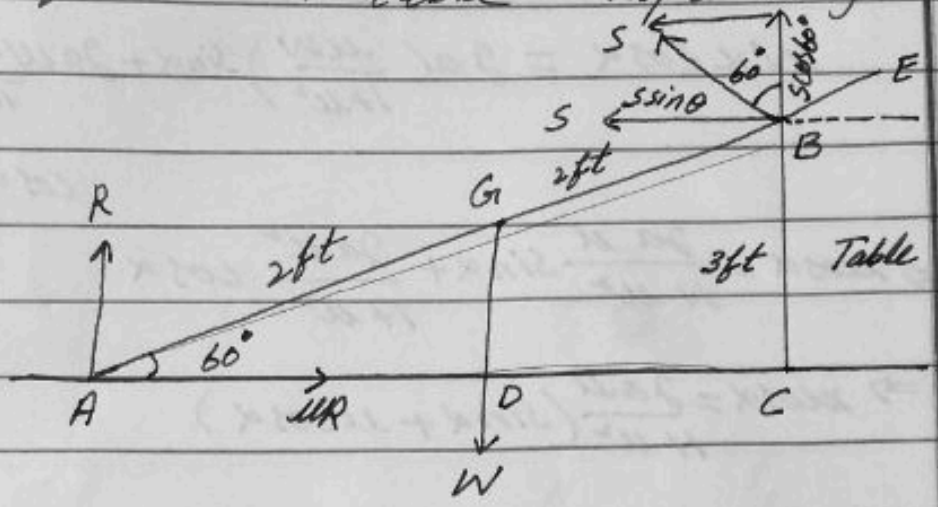
$$\Rightarrow x = \frac{2a\mu}{1+\mu^2}(\tan\alpha + \mu)$$

Ans

Q:- "A rod 4ft long rest on a rought floor against the smooth edge of a table of height 3ft. If the rod is on the point of slipping when inclined plane at an angle 60° to the horizental find the co-efficient of friction."

Sol:- Let μ be the co-efficient of friction. Let AE be the rod with mid point 'G'. Let W be the weight of the rod. Let R and S be the normal reaction

of floor and table respectively



Re-solving the forces horizantally and vertically.

$$\mu R = S \sin 60^\circ$$

$$\text{or } \mu R = S \cdot \frac{\sqrt{3}}{2}$$

$$\text{OR } R = \frac{S \cdot \sqrt{3}}{2\mu} \quad \text{—— (1)}$$

$$R + S \sin 60^\circ = W \quad \text{—— (2)}$$

$$R + S\frac{1}{2} = W \quad \text{—— (2)}$$

Taking moment about A.

$$W \cdot AD = S \cdot AB$$

In right $\triangle AGD$

$$\cos 60^\circ = \frac{AD}{AG} \Rightarrow \frac{1}{2} = \frac{AD}{2} \Rightarrow AD = 1$$

In right $\triangle ABC$

$$\sin 60^\circ = \frac{BC}{AB} \Rightarrow \frac{\sqrt{3}}{2} = \frac{3}{AB} \Rightarrow AB = \frac{6}{\sqrt{3}}$$

Put in (3)

$$W \cdot 1 = S \cdot \frac{6}{\sqrt{3}}$$

$$W = \frac{6S}{\sqrt{3}} \quad \text{---} \quad (4)$$

Eliminating R from (1) and (2)

$$\frac{S\sqrt{3}}{2\mu} + \frac{S}{2} = W$$

$$\Rightarrow S\left(\frac{\sqrt{3}}{2\mu} + \frac{1}{2}\right) = W$$

$$\Rightarrow S\left(\frac{\sqrt{3} + \mu}{2\mu}\right) = W$$

$$\Rightarrow S = \frac{2\mu W}{\sqrt{3} + \mu}$$

Put in eq (4)

$$W = \frac{6}{\sqrt{3}}\left(\frac{2\mu W}{\sqrt{3} + \mu}\right) \Rightarrow 1 = \frac{12\mu}{\sqrt{3}(\sqrt{3} + \mu)}$$

$$\Rightarrow 12\mu = \sqrt{3}(\sqrt{3} + \mu)$$

$$\Rightarrow 12\mu = 3 + \sqrt{3}\mu$$

$$12\mu - \sqrt{3}\mu = 3$$

$$(12 - \sqrt{3})\mu = 3 \Rightarrow \mu = \frac{3}{12 - \sqrt{3}}$$

$$\mu = 0.292 \quad \text{Ans}$$

Put in (3)

$$W \cdot 1 = S \cdot \frac{6}{\sqrt{3}}$$

$$W = \frac{6S}{\sqrt{3}} \quad \text{---} \quad (4)$$

Eliminating R from (1) and (2)

$$\frac{S\cdot\sqrt{3}}{2\mu} + \frac{S}{2} = W$$

$$\Rightarrow S\left(\frac{\sqrt{3}}{2\mu} + \frac{1}{2}\right) = W$$

$$\Rightarrow S\left(\frac{\sqrt{3}+\mu}{2\mu}\right) = W$$

$$\Rightarrow S = \frac{2\mu W}{\sqrt{3}+\mu}$$

Put in eq (4)

$$W = \frac{6}{\sqrt{3}}\left(\frac{2\mu W}{\sqrt{3}+\mu}\right) \Rightarrow 1 = \frac{12\mu}{\sqrt{3}(\sqrt{3}+\mu)}$$

$$\Rightarrow 12\mu = \sqrt{3}(\sqrt{3}+\mu)$$

$$\Rightarrow 12\mu = 3 + \sqrt{3}\mu$$

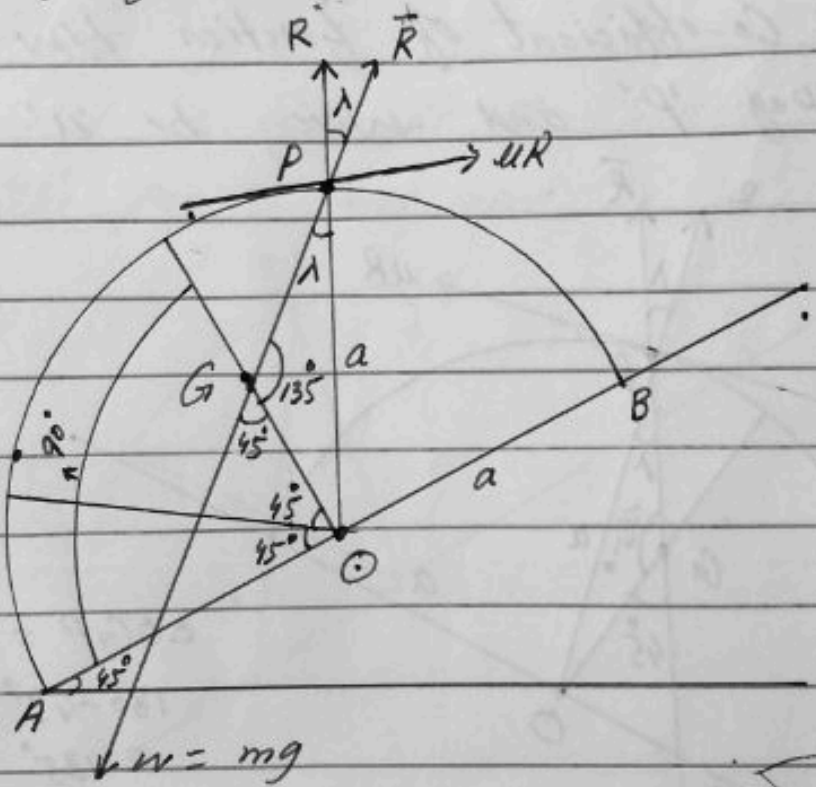
$$12\mu - \sqrt{3}\mu = 3$$

$$(12-\sqrt{3})\mu = 3 \Rightarrow \mu = \frac{3}{12-\sqrt{3}}$$

$$\mu = 0.292 \quad \text{Ans}$$

(i) A uniform semicircular wire hangs on a rough peg. The line joining its extremities making an angle of 45°, with the horizontal. If it is just on the point sliping find the co-efficient of friction b/w the wire and peg?

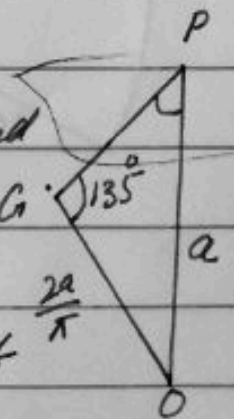
Sol:- Let 'u' be the co-efficient of friction b/w Peg 'P' and wire



Let 'O' be the Centre and a the radius of the semi-circular wire.

Let 'G' be the centre of gravity of the semicircular wire

Then we know. $OG = \frac{2a}{\pi}$ ⇒ formula



Let 'R' be normal reaction of peg 'P' Let $\bar{R}$ be the resultant reaction of R and μR.

Let 'λ' be the angle of friction.

"Applying the Law of sine $\triangle OPG_1$"

$$\frac{OG_1}{\sin\lambda} = \frac{OP}{\sin 135^\circ}$$

$$\frac{\frac{2a}{\pi}}{\sin\lambda} = \frac{a}{\frac{1}{\sqrt{2}}}$$

$\sin(135^\circ) = \sin(90^\circ + 45^\circ) = +\cos 45^\circ = \frac{1}{\sqrt{2}}$

$$\Rightarrow \frac{2a}{\pi \sin\lambda} = \frac{a}{\sqrt{2}} \Rightarrow \sin\lambda = \frac{\sqrt{2}}{\pi}$$

Now//

$$\cos\lambda = \sqrt{1 - \sin^2\lambda}$$

$$// = \sqrt{1 - \left(\frac{\sqrt{2}}{\pi}A\right)^2}$$

$$// = \sqrt{1 - \frac{2}{\pi^2}}$$

$$\cos\lambda = \sqrt{\frac{\pi^2 - 2}{\pi^2}} = \frac{\sqrt{\pi^2 - 2}}{\pi}$$

Since:-

$$\tan\lambda = \frac{\sin\lambda}{\cos\lambda}$$

$$\mu = \frac{\sqrt{2}}{\pi} \times \frac{\pi}{\sqrt{\pi^2 - 2}} = \sqrt{\frac{2}{\pi^2 - 2}}$$

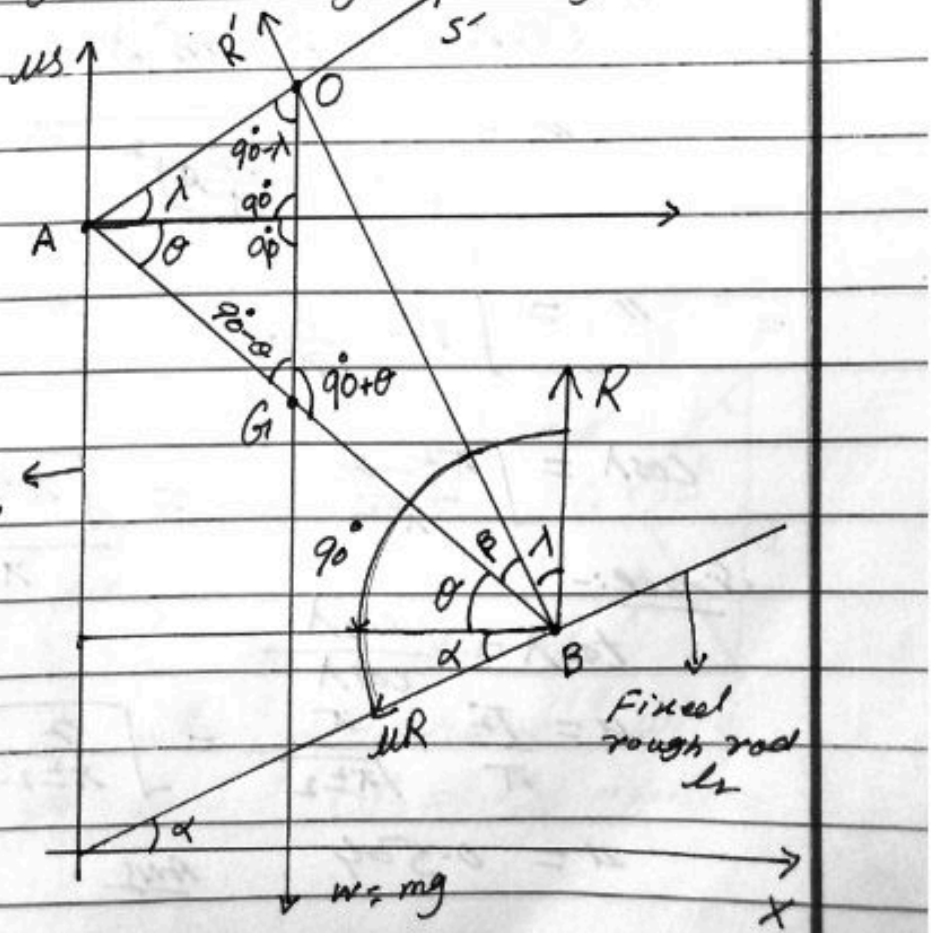
$$\mu = 0.504 \quad \text{ANS}$$

Q:- A uniform rod slides with its ends on two fixed equally rough rods one being vertical and other inclined at an angle α to the horizontal. Show that the angle θ to the horizontal of the moveable rod when it is on the point of sliding is given by

$$\tan\theta = \frac{1-2\mu\tan\alpha-\mu^2}{2(\mu+\tan\alpha)}$$

Sol:

Consider a uniform rod AB of lenght $2a$ and weight $W=mg$



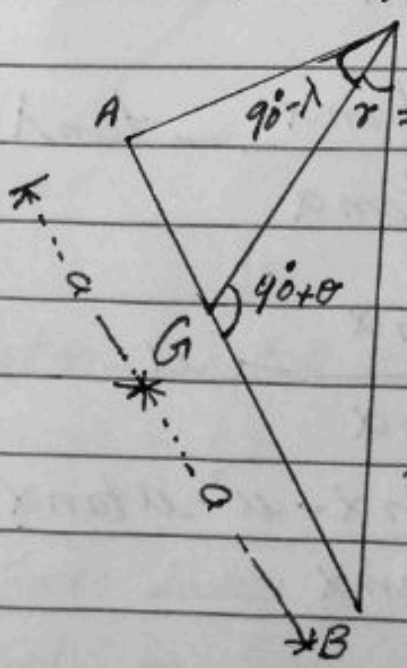
slides with its ends on two fixed equally rough rods l_1 and l_2. The rod l_1 is vertical and the rod l_2 is inclined at an angle α with horizontal.
Let 'θ' be the angle of the movable rod AB with the horizental. When it is on the point of sliding To prove.

$$\tan\theta = \frac{1 - 2\mu\tan\alpha - \mu^2}{2(\mu + \tan\alpha)}$$

The various forces acting on the system are show in fig.

Since: $\lambda + \beta + \theta + \alpha = 90^\circ$

so $\beta = 90^\circ - \lambda - \theta - \alpha$



from triangle OGB

$r + (90^\circ + \theta) + \beta = 180^\circ$

$r(90^\circ + \theta) + (90^\circ - \lambda - \theta - \alpha)$

$= 180^\circ$

$\gamma + 90^\circ + \theta + 90^\circ - \lambda - \theta - \alpha = 180^\circ$

$\gamma = \lambda + \alpha$

Applying (m·n) Theorm to the triangle OAB

$(a+a)\cot(90^\circ+\theta) = a\cot(90^\circ-\lambda) - a\cot(\lambda+\alpha)$

$2a(-\tan\theta) = a\tan\lambda - a\cot(\lambda+\alpha)$

$2(-\tan\theta) = -\cot(\lambda+\alpha) + \tan\lambda$

$2\tan\theta = \cot(\lambda+\alpha) - \tan\lambda$

$= \frac{1}{\tan(\lambda+\alpha)} - \tan\lambda$

$= \frac{1}{\frac{\tan\lambda + \tan\alpha}{1 - \tan\lambda\tan\alpha}} - \tan\lambda$

$2\tan\theta = \frac{1 - \tan\lambda\tan\alpha}{\tan\lambda + \tan\alpha} - \tan\lambda$

$2\tan\theta = \frac{1 - \mu\tan\alpha}{\mu + \tan\alpha} - \mu$

$2\tan\theta = \frac{1 - \mu\tan\alpha - \mu^2 - \mu\tan\alpha}{\mu + \tan\alpha}$

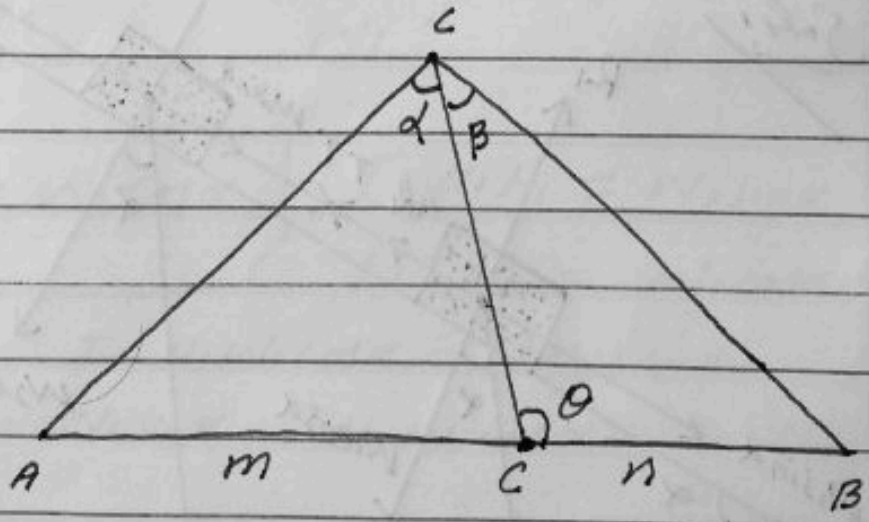
$$2\tan\theta = \frac{1 - 2\mu\tan\alpha - \mu^2}{\mu + \tan\alpha}$$

$$\tan\theta = \frac{1 - 2\mu\tan\alpha - \mu^2}{2(\mu + \tan\alpha)}$$

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(1) If a body is in equilibrium under the action three forces they are either Concurrent or Parallel

(2) (m, n) Theorm :-



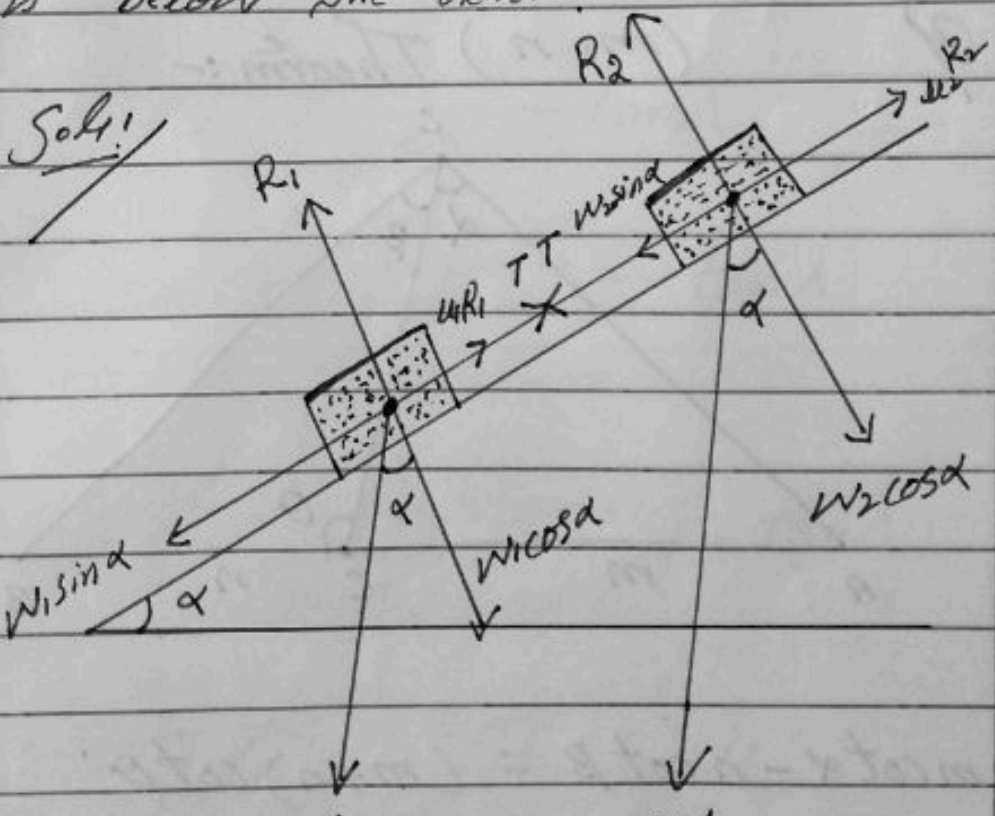
$$m\cot\alpha - n\cot\beta = (m+n)\cot\theta$$

Q :-

Two bodies of weight W_1, W_2 are placed on a rough inclined plane

and are Connecting by a light string which Coincides with a line of greatest slople of The plane - If the Co-efficient of friction b/w The bodies and The plane be resp. μ_1 & μ_2 find The in-clination of The plane to The horizental. When both bodiesare on The point of motion it being assumed That The smooth body is below The other?

Sol:-



$W_1 = m_1 g$ $W_2 = m_2 g$

Consider two bodies of weigh

W_1 & W_2 are placed on an inclined plane whose inclined/inclination with the horizontal is 'α'
The bodies are connected by light string and the co-efficient of friction b/w the bodies and the plane be resp μ_1 and μ_2
Let "T" be the tension of in string
The various forces acting on the ~~body~~ of system are show in fig.

Equilibriam of the body of weight W_1 "

$R_1 = W_1 \cos\alpha$

$T + \mu_1 R_1 = W_1 \sin\alpha$

Put $R_1 = W_1 \cos\alpha$

$T + \mu_1 W_1 \cos\alpha = W_1 \sin\alpha$

$T = W_1 \sin\alpha - \mu_1 W_1 \cos\alpha$ —— (1)

Equilibriam of the body of weight W_2 "

$R_2 = W_2 \cos\alpha$

$T + W_2 \sin\alpha = \mu_2 R_2$

Put $R_2 = W_2 \cos\alpha$

$$T = \mu_2 W_2 \cos\alpha - W_2 \sin\alpha$$

Equating (1) & (2)

$$W_1 \sin\alpha - \mu_1 W_1 \cos\alpha = \mu_2 W_2 \cos\alpha - W_2 \sin\alpha$$

$$W_1 \sin\alpha + W_2 \sin\alpha = \mu_1 W_1 \cos\alpha + \mu_2 W_2 \cos\alpha$$

$$(W_1 + W_2)\sin\alpha = \cos\alpha(\mu_1 W_1 + \mu_2 W_2)$$

$$\Rightarrow \frac{\sin\alpha}{\cos\alpha} = \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}$$

$$\tan\alpha = \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}$$

$$\alpha = \tan^{-1}\left(\frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2}\right)$$

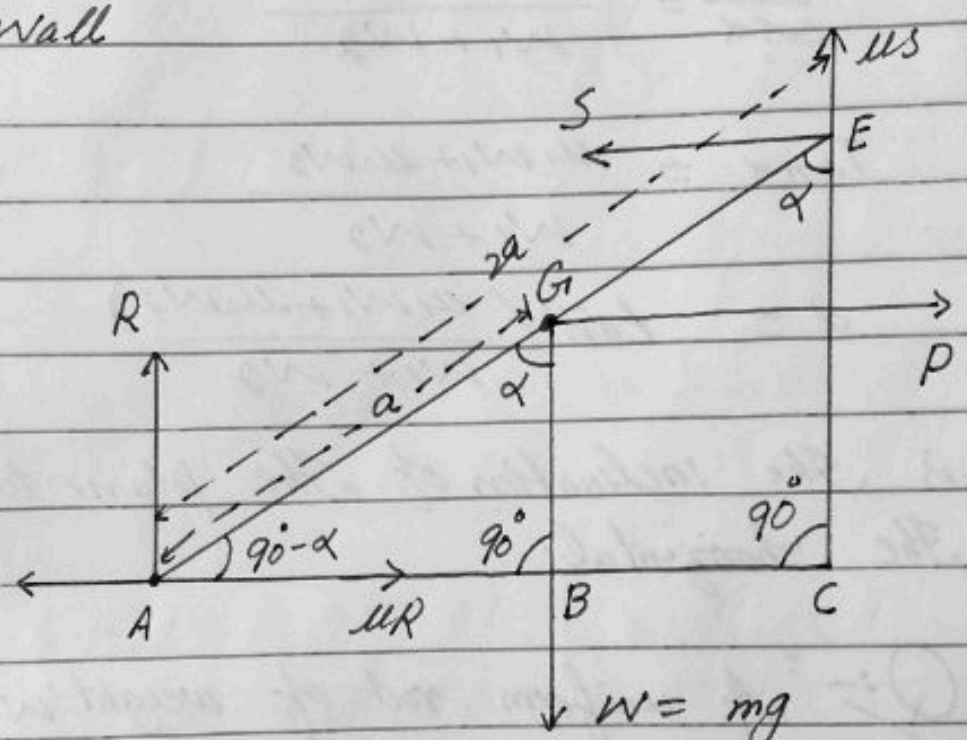
is the inclination of the plane to the horizental.

Q:- A uniform rod of weight 'W' is placed with its lower end on a rough horizental floor and upper end against an equally rough vertical wall. The rod makes an angle 'α' with the wall and is just prevented from slipping down by a horizental force 'P' applied at its middle point. Prove that :-

$$P = W\tan(\alpha - 2\lambda)$$

Where 'λ' is the angle of friction and $\lambda < \frac{1}{2}\alpha$.

Soln:- Let '2a' be the lenght of the rod 'AE'. Let R and 'S' be the normal reaction of floor and wall



Let 'μ' be the Co-efficient of friction b/w rod and floor and rod and wall.

Resolving the forces horizontal and vertically. "we get"

$$R + \mu S = W \quad \text{---(1)}$$

$$P + \mu R = S \quad \text{---(2)}$$

Taking moments about 'A'

$W \cdot AB + P \cdot DC = S \cdot CE + 2S \cdot AC$ — (3)

In right 'ABG'

$\cos(90^\circ - \alpha) = \frac{AB}{AG} \Rightarrow \sin\alpha = \frac{AB}{a}$

$\Rightarrow AB = a\sin\alpha$

In right 'ΔABG'

$\sin(90^\circ - \alpha) = \frac{BG}{AG} \Rightarrow \cos\alpha = \frac{DC}{a}$

$\Rightarrow DC = a\cos\alpha$

In right Δ ACE'

$\sin(90^\circ - \alpha) = \frac{CE}{AE} \Rightarrow \cos\alpha = \frac{CE}{2a}$

$\Rightarrow CE = 2a\cos\alpha$

In right ΔACE'

$\cos(90^\circ - \alpha) = \frac{AC}{AE} \Rightarrow \sin\alpha = \frac{AC}{2a}$

$\Rightarrow AC = 2a\sin\alpha$

Put in (3) All these values

$W \cdot a\sin\alpha + P a\cos\alpha = S \cdot 2a\cos\alpha + 2aS \cdot \sin\alpha$

$\Rightarrow W\sin\alpha + P\cos\alpha = 2S\cos\alpha + 2S\sin\alpha$ — (4)

Eliminating 'R' from (1) and (2)

$$P + \mu(W - \mu S) = S$$

$$\Rightarrow P + \mu W - \mu^2 S = S$$

$$\Rightarrow P + \mu W = S + \mu^2 S$$

$$\Rightarrow P + \mu W = S(1 + \mu^2)$$

$$\Rightarrow S = \frac{P + \mu W}{1 + \mu^2}$$

Put in (4)

$$W\sin\alpha + P\cos\alpha = 2\left(\frac{P+\mu W}{1+\mu^2}\right)\cos\alpha + 2\mu\left(\frac{P+\mu W}{1+\mu^2}\right)\sin\alpha$$

$$\Rightarrow (1+\mu^2)W\sin\alpha + (1+\mu^2)P\cos\alpha = 2(P+\mu W)\cos\alpha + 2\mu(P+\mu W)\sin\alpha$$

$$\Rightarrow (1+\mu^2)W\sin\alpha + (1+\mu^2)P\cos\alpha = 2P\cos\alpha + 2\mu W\cos\alpha + 2\mu P\sin\alpha + 2\mu^2 W\sin\alpha$$

$$\Rightarrow P(\cos\alpha + \mu^2\cos\alpha - 2\cos\alpha - 2\mu\sin\alpha) = W(2\mu\cos\alpha + 2\mu^2\sin\alpha - \mu^2\sin\alpha)$$

$$\Rightarrow P = \frac{W(2\mu\cos\alpha + 2\mu^2\sin\alpha - \sin\alpha - \mu^2\sin\alpha)}{(-2\mu\sin\alpha + \mu^2\cos\alpha - \cos\alpha)}$$

$$= \frac{W(2\mu\cos\alpha - (1-\mu^2)\sin\alpha)}{-(2\mu\sin\alpha - \mu^2\cos\alpha + \cos\alpha)}$$

Dividing numerator and denominator by $(1-\mu^2)\cos\alpha$

$$P = \frac{-W\left(\frac{2\mu}{1-\mu^2} - \tan\alpha\right)}{\left(\frac{2\mu}{1-\mu^2}\tan\alpha + 1\right)}$$

$$P = \frac{-W\left(\frac{2\tan\lambda}{1-\tan^2\lambda} - \tan\alpha\right)}{\left(\frac{2\tan\lambda}{1-\tan^2\lambda}\tan\alpha + 1\right)}$$

$$= \frac{W\left(\tan\alpha - \frac{\tan\lambda+\tan\lambda}{1-\tan\lambda\tan\lambda}\right)}{\left(\frac{\tan\lambda+\tan\lambda}{1-\tan\lambda\cdot\tan\lambda}\cdot\tan\alpha + 1\right)}$$

$$= \frac{W(\tan\alpha - \tan(\lambda+\lambda))}{(\tan(\lambda+\lambda)\tan\alpha + 1)}$$

$$= \frac{W(\tan\alpha - \tan 2\lambda)}{1 + \tan 2\lambda\cdot\tan\alpha}$$

$$P = W\tan(\alpha - 2\lambda)$$

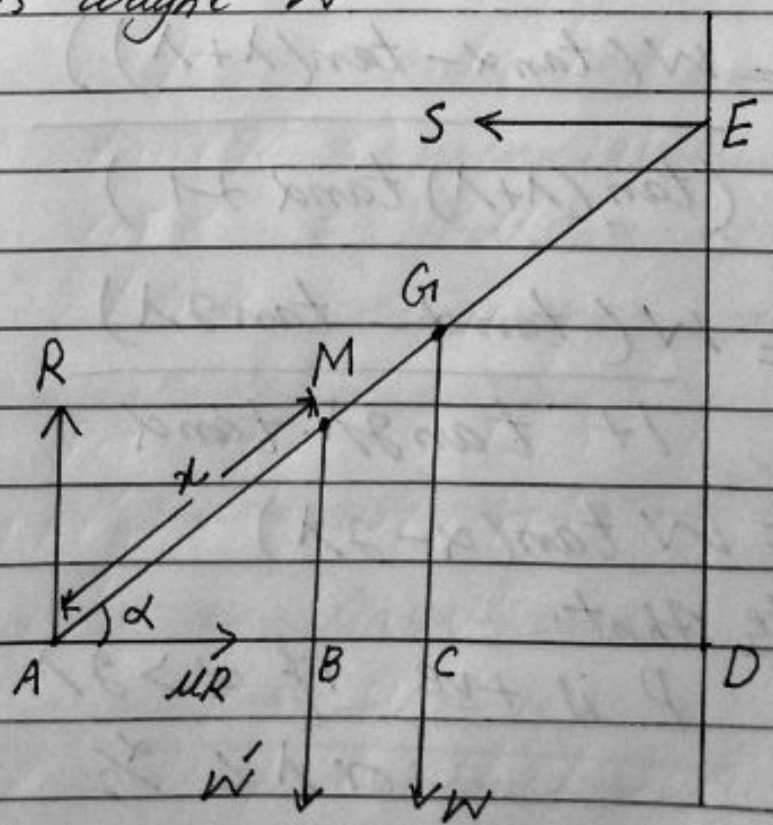
Note That.

P is +ve if $\alpha > 2\lambda$

or $\lambda < \alpha/2$

Q:- A uniform ladder rests in limiting equilibrium with one end on rought horizental plane and the other against a smooth vertical wall. A man ascends the ladder. Show that he can-not go more than Half way up.

Sol: Let 'x' the distance, when of the man 'M' when the ladder slips.
Let 2a be lenght and W be the weight of the ladder 'AE', Let man's weight 'W'.



Let 'R' and 'S' be the normal Reaction of floor and wall respt-ly. Let 'μ' be the co-efficient of friction b/w the ladder and floor.

Resolving the forces Horizental and vertically, we get

$$R = W + W' \quad \text{---} (1)$$

$$S = \mu R \quad \text{---} (2)$$

Taking moments about 'A'

$$W \cdot AC + W' \cdot AB = S \cdot DE \quad \text{---} (3)$$

In right '$\triangle ACG$'

$$\cos\alpha = \frac{AC}{AG} \Rightarrow AC = a\cos\alpha$$

In right '$\triangle ABM$'

$$\cos\alpha = \frac{AB}{AM} \Rightarrow AB = x\cos\alpha$$

In right '$\triangle ADE$'

$$\sin\alpha = \frac{DE}{AE} \Rightarrow \sin\alpha = \frac{DE}{2a} \Rightarrow DE = 2a\sin\alpha$$

Put in eq (3) we get

$W \cdot a\cos\alpha + W' \cdot x\cos\alpha = S \cdot 2a\sin\alpha$ (4)

Eliminating R from (1) & (2)

$$S = \mu(W+W')$$

$Wa\cos\alpha + W'x\cos\alpha = 2a\mu(W+W')\sin\alpha$

$\Rightarrow Wa + W'x = 2a\mu(W+W')\tan\alpha$ — (5)

We Know That,

Angle of ladder to vertical $= \tan^{-1}(2\mu)$

In This case The ladder is in limiting equilibrium and ~~limiting~~ inclination of The ladder to The vertical is $(90^\circ - \alpha)$

$$\therefore 90 - \alpha = \tan^{-1}(2\mu)$$

$$\Rightarrow \tan(90^\circ - \alpha) = 2\mu$$

$$\Rightarrow \cot\alpha = 2\mu$$

$$\Rightarrow \tan\alpha = \frac{1}{2\mu}$$

Put in eq (5)

$$Wa + W'x = 2a\mu(W+W') \cdot \frac{1}{2\mu}$$

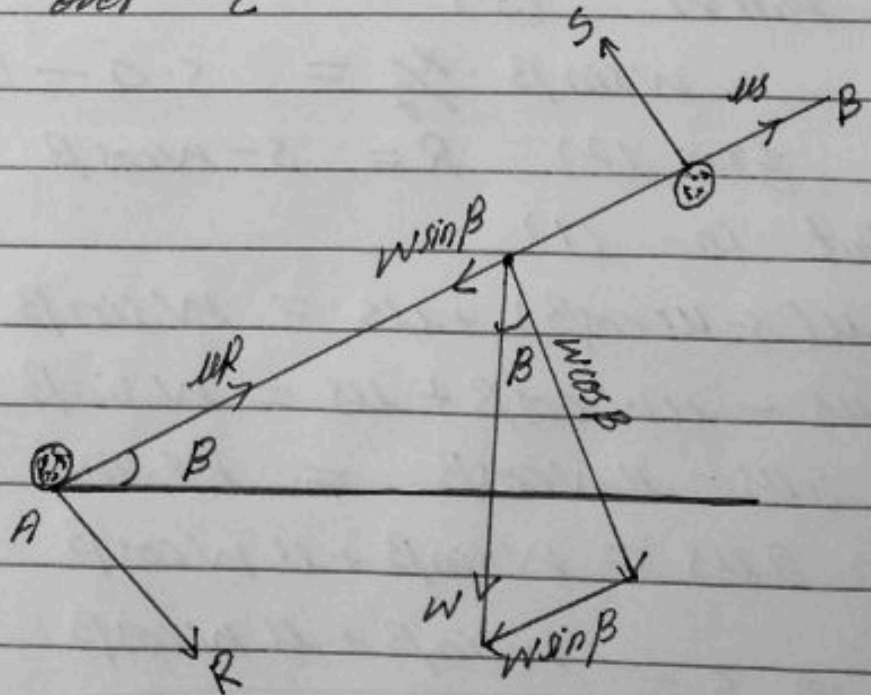
$$\Rightarrow W'x = aW'$$

$$\Rightarrow x = a$$

x = half lenght of The ladder.

Q:- A thin uni-form rod passes over one peg and under another The co-efficient of friction b/w each peg and the rod being 'u' The distance b/w the peg is 'a' and the straight line joining them makes an angle 'β' with the Horizontal. Show that equi-librium is not possible unless the lenght of the rod is greater than $\frac{a}{u}(u + \tan\beta)$

Sol:- Suppose a rod 'AB' of lenght 'x' passes through a peg 'A' and over 'C'



Let 'R' and 'S' be the normal reactions of Peg 'A' and Peg 'C'

Let 'w' be the weight of rod
Let u be the co-efficient of friction
Resolving the forces Horizontal and vertical. we get

$uR + uS = W\sin\beta$ —— (1)

$S = R + W\cos\beta$ —— (2)

Taking moments about 'A'

$W\cos\beta \cdot AG = S \cdot AC$ —— (3)

$\because$ But $AG = x/2$

$uR + uS = W\sin\beta$ —— (4)

$W\cos\beta / AG = S / AC$ —— (3)

$AC = a$

Putin (3)

$W\sin\beta \cdot x/2 = S \cdot a$ — (5)

from (2) $R = S - W\cos\beta$

Put in (1)

$u(S - u\cos\beta) + uS = W\sin\beta$

$uS - uW\cos\beta + uS = W\sin\beta$

$2uS - uW\cos\beta = W\sin\beta$

$\Rightarrow 2uS = W\sin\beta + uW\cos\beta$

$\Rightarrow S = \dfrac{W\sin\beta + uW\cos\beta}{2u}$

Put in eq (4)

$$W\cos\beta\,\frac{x}{g} = \frac{W(\sin\beta+\mu\cos\beta)}{g\mu}\cdot a$$

$$x\cos\beta = \frac{a}{\mu}(\sin\beta+\mu\cos\beta)$$

$$\Rightarrow x = \frac{a}{\mu}(\tan\beta+\mu)$$ Ans